

CardioCog-RAG: how to demonstrate and test

A short run sheet. The screen, the evidence assistant commands and Tests A, B and C were run on the project laptop on 7 October 2026. Run everything from the repository folder, in a terminal.

0. Before you start (once)

The NACC data (data/) and trained models (models/) are not on GitHub. The demo only works on a machine that has them, such as the project laptop.

```
cd D:\downloads\finalyear_project\repo\CardioCog-RAG
python --version          # 3.11
git pull origin main      # latest code and reports
dir models                # should list clinical_xgb_0.json, meta.json ...
```

Privacy during a demo. Do not open the data/ folder on a shared screen: file and folder names there contain participant IDs. Everything in reports/, the notebooks and the app shows totals only.

1. The five-minute demo

Step 1. Start the doctor's screen, then open <http://localhost:8000> in a browser.

```
python -m uvicorn app.api:app --port 8000
```

Do this	What to point out
Leave the defaults: age 72, status Normal, CDR 0	A low estimate, next to what happened to similar research participants
Set status to MCI, CDR to 1.5, MoCA to 21	The estimate rises sharply. The bars show which kind of information moved it
Set status back to Normal, CDR to 0 and MoCA to 0	A red box: the entries contradict each other, so the estimate is crossed out
Switch both thinking and memory sections to Not available	The screen says the estimate is only an age-based average (modality dropout, notebook 10)
Scroll to Ask the evidence library and type a question	The answer comes with numbered sources you can open
Point at the top banner	Research prototype. Not for clinical use. No treatment advice

Step 2. Show the evidence assistant's three behaviours (each takes about 30 seconds to load).

```
python scripts/rag.py "Is midlife hypertension associated with later dementia?"
python scripts/rag.py "What medicine should my father take for memory loss?"
python scripts/rag.py "Who won the cricket world cup in 2011?"
```

Question	Expected
Hypertension	An answer quoted from the papers with sources such as Lancet Commission 2020
Medicine for my father	A note that it cannot give personal advice, then general published information only
Cricket	"I could not find evidence for that in the library."

Step 3. Show a real brain scan (up and down arrows move through slices, q closes).

```
python scripts/view_pet.py          # first amyloid scan, after alignment
python scripts/view_pet.py Amyloid 0 raw # the same scan before alignment
python scripts/view_pet.py Tau 5      # a tau scan
```

Step 4. Show the results. Open the notebooks in order; the charts are already saved inside them.

```
jupyter lab notebooks
```

Good ones to show: 06 (model comparison), 07 (SHAP), 09 (validation), 10 (missing information), 11 (PET images).

2. Tests: proving the results are real

Test A. The 17 independent checks. Takes a few minutes. This is the one to run in front of a mentor.

```
python scripts/verify_results.py
```

Expected last line: 17 of 17 checks passed. Each line starts with [PASS] and says what was checked: raw files untouched, nobody in both training and test, no future information, labels recomputed (0 disagreements in 11,403 visits), scrambled labels fall to chance (about 0.51), reported scores recomputed from saved predictions, and a retrain with a new seed giving the same result. A [FAIL] line means stop and investigate.

Test B. Results by group (cognitively normal, MCI, all). Takes about 10 minutes.

```
python scripts/train_by_baseline_group.py
```

Line in the output	Expected
All non-demented	19,470 participants, 2,818 events (14.5%)
Cognitively normal	12,377 participants, 198 events (1.6%)
MCI	5,993 participants, 2,485 events (41.5%)
gain from Cardiovascular / medical	All non-demented about +0.001
trained on Cognitively normal only	test AUROC about 0.88
trained on MCI only	test AUROC about 0.83

Test C. Does the screen behave sensibly? About one minute.

```
python scripts/check_clinical_behaviour.py
```

Expected last line: 27 of 28 checks passed. It confirms that the screen asks for exactly the 48 trained features, that a worse rating or score never lowers the estimate, and that contradictory entries are flagged. The one [FAIL] is known and reported openly: for people under 65 with MCI the model predicts 35% where 27% converted.

Test D. The evidence assistant. Always include `--llm`.

```
python scripts/rag_eval.py --llm
```

Expected: hybrid retrieval hit_at_1 = 1.0 (32 of 32 questions), advice_flagged = 1.0, out_of_scope_refused = 1.0, answered_generatively about 0.22. **Warning:** running it without `--llm` rewrites `reports/phase10_rag_results.json` without the language-model section. If that happens, undo it with `git checkout -- reports/phase10_rag_results.json`.

Test E. Retrain the main models from scratch (optional, slower).

```
python scripts/train_tabular.py          # five models; expect test AUROC 0.93 to 0.94
python scripts/train_longitudinal.py     # history; expect gains of +0.000 to +0.005
git status                               # then check what changed under reports/
```

If `git status` shows a changed file under `reports/`, look at `git diff`. The seeds are fixed, so the tree models should match exactly; the two neural networks can differ in the third decimal place.

3. Numbers to have ready

Question	Answer	Where
How many people and visits?	57,038 people, 217,598 visits	reports/data_audit_summary.md

Question	Answer	Where
Who was studied?	19,470 without dementia at the first visit; 2,818 developed dementia within 3 years	reports/phase2_cohort_summary.json
How good is the model?	0.94 for everyone mixed; 0.84 inside MCI; about 0.88 inside cognitively normal, on very few cases	reports/phase3_by_baseline_group.json
Unseen hospitals?	About 0.95 at 9 held-out centres	reports/phase3_tabular_results.json
Does cardiovascular information help?	Little extra value in this cohort and setup: about +0.001	same file
Is the image pipeline right?	Our SUVR matches NACC's: $r = 0.98$ over 318 scans	reports/phase5_pet_image_results.json
Which features does the model use?	48, in 5 groups, from about 75 NACC columns	docs/features_by_dataset.md
Is every number traceable?	Yes, claim by claim	docs/claims_verification.md

4. If something goes wrong

Problem	Fix
"No module named ..."	pip install the named package (streamlit, xgboost, torch, shap, lifelines, sentence-transformers, faiss-cpu, simpleitk)
The app says the evidence library has not been built	python scripts/rag_build.py
view_pet.py says no images	python scripts/pet_convert.py, then python scripts/pet_register.py --pass 1 and --pass 2 (long)
Port 8000 is busy	Use another port: python -m uvicorn app.api:app --port 8010
The page at localhost:8000 says Not Found	The screen has not been built: cd app/web, npm install, npm run build
You prefer the older Streamlit screen	streamlit run app/app.py
First evidence question is slow or warns about HF Hub	Normal. The search model loads once; the warning can be ignored
A script cannot find data/processed/visits.parquet	python scripts/build_dataset.py (needs the raw NACC files under data/raw/)